# Security Evaluation of Face Biometric Templates: Data Engineering and a FaceNet Verification Baseline

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***Abstract*—This checkpoint establishes a reproducible, unprotected face-template verification baseline for a larger study of embedding inversion and cancelable biometric protection. We use a filtered Labeled Faces in the Wild (LFW) dataset with 4,324 images from 158 identities and a pretrained InceptionResnetV1 face encoder. At a threshold selected on validation identities, the held-out test evaluation yields 7 false acceptances in 1,500 impostor trials (FAR 0.47%) and 75 false rejections in 1,500 genuine trials (FRR 5.00%). The threshold-independent test EER is approximately 2.30% and ROC-AUC is 0.99696. No inversion or template-protection claims are made at this checkpoint.**

**Nature of project:** Test inversion vulnerability of face templates and evaluate cancelable protection; this checkpoint builds the unprotected verification baseline.

## I. Problem Statement and Motivation

Face recognition systems frequently store feature vectors (embeddings) derived from a user's face rather than only a photograph. Such vectors enable similarity-based identity verification but may still retain identity-sensitive information. Unlike passwords, the underlying face characteristic cannot simply be replaced following a template breach. Our larger project asks whether a revocable transformation can reduce the identity information recoverable from compromised templates without substantially degrading legitimate verification. The proposed future comparison is raw-template inversion versus inversion or recovery attempts against protected templates under explicitly matched attacker assumptions. The present checkpoint is limited to dataset engineering, exploratory analysis, and the unprotected verification baseline.

**Scope clarification relative to proposal:** LFW is used as the initial primary benchmark instead of the proposed CelebA subset, to obtain a tractable and reproducible baseline in Google Colab. CelebA remains a possible later evaluation dataset. We use a pretrained network rather than training a recognition network from scratch. We do not claim that either inversion or cancelable protection has been implemented.

## II. Data Engineering

### A. Origin and Access

We use Labeled Faces in the Wild (LFW), a public research benchmark of unconstrained face photographs [1]. The data are fetched by scikit-learn's `fetch_lfw_people` loader; dataset access and reuse remain subject to the original LFW terms and image provenance. Our code uses `min_faces_per_person=10`, `resize=1.0`, and `color=True`. No dataset files are committed to the repository.

### B. Characteristics, Sanity Checks and Preprocessing

The resulting subset contains **4,324 images, 158 labeled identities**; the minimum, median, and maximum image counts per identity are **10, 17, and 530**, respectively. The loaded RGB crops are **125** $\times$ **94** $\times$ **3** with values in $[0, 1]$. The notebook checks that image and label counts agree, all pixels are finite, and there are no exact duplicate image arrays (**zero found**). This exact-array check cannot rule out near-duplicate photographs or identity-label errors. LFW has substantial class imbalance, and its images vary in illumination, pose, and resolution. Session identifiers and spoof/live labels are not available in this filtered dataset; the labels denote identity only.

For embedding extraction, images are converted to tensors, rescaled to the 0–255 range, bilinearly resized to $160 \times 160$, and standardized using $(x - 127.5)/128$. The pretrained `facenet-pytorch` InceptionResnetV1 model (VGGFace2 weights) outputs 512-dimensional vectors, which are L2-normalized before cosine comparison. We do not perform a separate landmark-based alignment stage: this is a limitation, not a claim of fully aligned faces. No PCA or feature selection is applied because the encoder provides fixed-size embeddings.

### C. Partitions and Trial Construction

With random seed **228**, the 158 identity labels are split into 79 validation and 79 test identities, with no overlap between these two groups. For each identity, one randomly selected image becomes its enrollment template and remaining images provide candidate genuine trials. Genuine trials are subsampled to 1,500 per partition. For every genuine trial, one impostor comparison is sampled using another enrolled identity's template, giving **1,500 genuine and 1,500 impostor trials** for each of validation and test. The validation partition selects the threshold; test labels are not used to choose it. These are evaluation identities, not an additional training split

TABLE I
OBSERVED DATASET AND TRIAL CHARACTERISTICS. COUNTS ARE FROM THE EXECUTED NOTEBOOK.

| Item | Observed value |
|---|---|
| Images / identities | 4,324 / 158 |
| Images per identity (min/median/max) | 10 / 17 / 530 |
| Exact duplicate image arrays | 0 |
| Image size / embedding size | $125 \times 94 \times 3$ / 512 |
| Validation identities / test identities | 79 / 79 |
| Genuine / impostor trials (per split) | 1,500 / 1,500 |

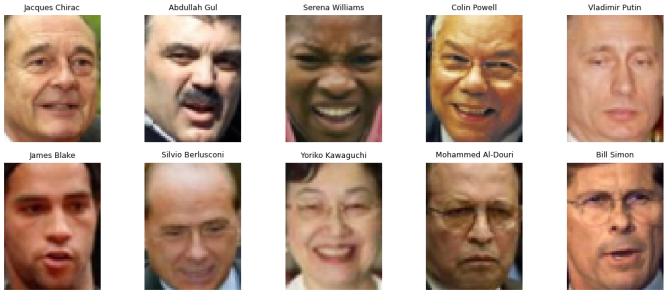



Fig. 1. Representative LFW face crops. Names are original dataset identity labels; these examples are not evidence of verification performance.

for the externally pretrained encoder. The model's external VGGFace2 training membership is not audited here.

## III. EXPLORATORY DATA ANALYSIS

The notebook generates `eda_univariate.png` with (i) images per identity, (ii) mean brightness per image, and (iii) image-contrast box plot; `eda_bivariate.png` plots brightness against contrast; and `eda_class_balance.png` displays the 20 most frequent identity labels. Each chart has a title and labeled axes. The count range of 10 to 530 confirms severe identity imbalance, motivating capped genuine-trial sampling and balanced impostor sampling. Brightness and contrast plots are intended to expose imaging variation that could affect matching, but no numerical brightness/contrast quantiles or correlations were preserved in the supplied result files. Therefore we do not assert trends or outlier counts beyond those actually logged. Figure 2 shows class imbalance and variation in brightness and contrast; Fig. 3 shows their joint distribution. The sample images in Fig. 1 illustrate variation in pose, expression, and crop quality.

### A. EDA Interpretation and Reporting Controls

The identities represented in LFW are not uniformly sampled from a population. Thus, an unweighted histogram of subjects and an image-weighted histogram answer different questions: the first describes subject diversity and the second describes exposure to images of frequent identities. For this checkpoint we report the observed distribution and deliberately balance trial labels, rather than claiming that balanced trials make the original image dataset balanced. The brightness–contrast scatter is a quality diagnostic, not evidence of a causal

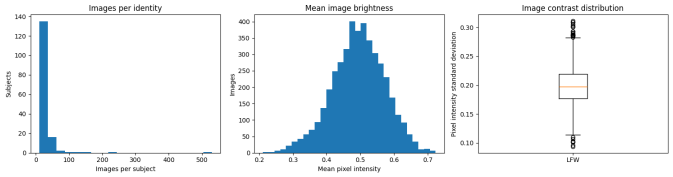



Fig. 2. Univariate EDA: images per subject, normalized mean brightness, and image contrast. The long-tailed subject counts motivate balanced verification trials.

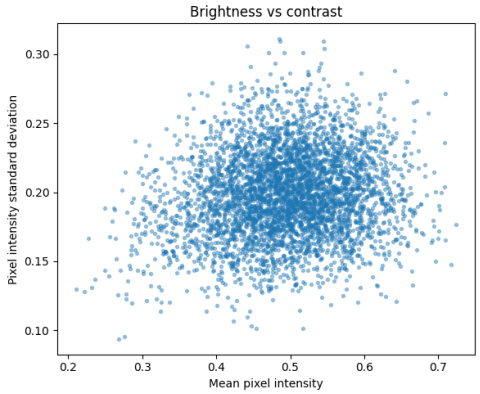



Fig. 3. Image brightness versus contrast. The spread reflects acquisition variation; it does not establish causality for verification errors.

relationship with verification errors. An additional planned analysis is to stratify genuine-pair similarity by brightness difference and to inspect whether low-quality crops cluster among false rejections.

### B. Data Quality Risks

Exact-duplicate checking found zero identical flattened image arrays. However, different crops of the same photograph, near-duplicate frames, and mislabeled identities would not be caught by this procedure. Because the dataset is composed of uncontrolled web photographs, pose, lighting, expression, and occlusion may vary within the same identity. The pretrained encoder's training set may also overlap with identities or source photographs in LFW; the present experiment is a practical verification baseline rather than a guarantee of strictly unseen pretraining identities.

## IV. BASELINE IMPLEMENTATION AND PRELIMINARY RESULTS

### A. Method and Threshold

For each trial, the normalized query embedding $q$ is compared with the normalized enrolled template $t$ using cosine similarity $s = q^\top t$. Accept if $s \geq \tau$, otherwise reject. We sweep validation similarity thresholds and select the threshold minimizing $|\mathrm{FAR} - \mathrm{FRR}|$. This yields $\tau = 0.48683995$ and validation FAR and FRR of **1.0% each**. This near-EER validation threshold is a convenient balanced-error operating point, not necessarily a deployment threshold: high-security deployments may instead target a lower FAR.

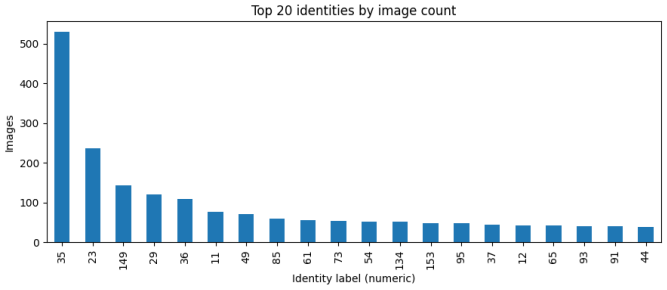



Fig. 4. Top 20 identities by sample count, showing concentration of images in a small number of identities.

TABLE II
BASELINE TEST METRICS AT THE VALIDATION-SELECTED THRESHOLD, EXCEPT EER AND AUC.

| Metric | Result |
|---|---|
| Threshold $\tau$ | 0.48684 |
| False accepts / impostor trials | 7 / 1,500 |
| False rejects / genuine trials | 75 / 1,500 |
| True accepts / true rejects | 1,425 / 1,493 |
| FAR / FRR | 0.47% / 5.00% |
| Test EER (threshold sweep) | 2.30% |
| Test ROC-AUC | 0.99696 |

### B. Held-Out Results

At the fixed validation-selected threshold, the test set yields **7 false acceptances / 1,500 impostor trials** and **75 false rejections / 1,500 genuine trials**. Consequently,

$$\mathrm{FAR} = \frac{7}{1500} = 0.00467, \quad \mathrm{FRR} = \frac{75}{1500} = 0.0500. \quad (1)$$

The resulting true acceptances and true rejections are 1,425 and 1,493, respectively. Sweeping the *test* scores for descriptive, threshold-independent assessment gives an approximate **EER of 2.30%** and **ROC-AUC of 0.99696**. The EER is not the error rate at the selected operating threshold; this distinction matters because the selected threshold produces unequal test FAR and FRR.

### C. Error Analysis, Reproducibility and Limitations

False rejections are more frequent than false acceptances at this operating point. Plausible causes include pose, lighting, crop quality, and a single enrollment image per subject; these are hypotheses and require inspection of individual failure cases before attribution. A required next analysis is to visualize rejected genuine pairs and accepted impostor pairs, with scores and image-quality measurements. The current evidence does not support claiming any particular subgroup is disproportionately affected.

The run fixes Python, NumPy, and PyTorch seeds at 228, disables cuDNN benchmarking, and enables deterministic cuDNN behavior. The supplied environment reports Python 3.13.16, PyTorch 2.11.0+cu130, torchvision 0.26.0+cu130, and scikit-learn 1.6.1. Reproducibility additionally depends on pinning packages, recording the pretrained weights version, and committing the executed notebook at a specific Git revision. **Repository URL:** `github.com/chinnikrishna854/CS228-Face-Biometric-Template-Security`. The repository must include dataset-fetch instructions, EDA and evaluation code, requirements, and the saved result metrics; it must not contain the dataset.

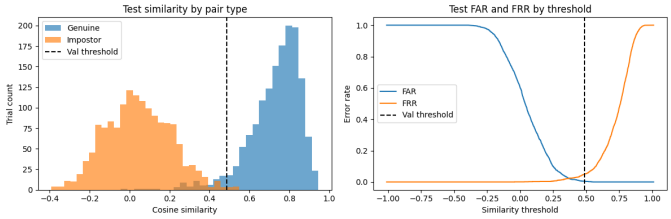



Fig. 5. Test genuine/impostor cosine similarities (left) and FAR/FRR versus threshold (right). Dashed lines show the threshold selected using validation identities only.

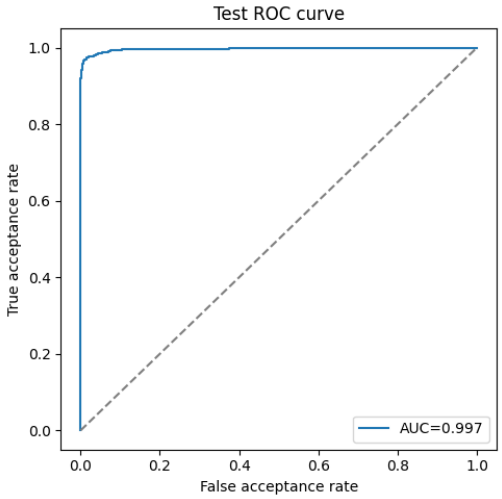



Fig. 6. Held-out test ROC curve; AUC = 0.99696. This is baseline verification performance, not inversion-attack performance.

Future stages will test optimization-based and learning-based template inversion and compare raw embeddings with keyed cancelable transformations. Protected-template attacks must use a clearly defined attacker knowledge/key-compromise model; direct reuse of a raw-embedding inverter on transformed vectors is not a valid like-for-like comparison.

### D. Next Checkpoint Boundary

The present baseline answers whether the unprotected embeddings support meaningful verification; it does not answer whether they can be inverted. The next phase will define the inversion threat model, develop a reconstruction method using separate training and evaluation identities, and evaluate identity leakage through an independent verifier. After that, keyed template transformations will be assessed for verification accuracy, inversion resistance, cross-key linkability, and revocability. All comparisons will be made under matched trial construction and explicitly stated assumptions about attacker access to the encoder and protection keys.

## REFERENCES


[1] G. B. Huang, M. Ramesh, T. Berg, and E. Learned-Miller, "Labeled Faces in the Wild: A Database for Studying Face Recognition in Unconstrained Environments," UMass Amherst Tech. Rep. 07-49, 2007.

[2] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A Unified Embedding for Face Recognition and Clustering," CVPR, 2015.

[3] V. Krivokuca Hahn and S. Marcel, "Towards Protecting Face Embeddings in Mobile Face Verification Scenarios," IEEE T-BIOM, 2022.